Energy Systems - Group Project

Group O

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1 Task 1 - Questions on primal solution

(a) Compute the annuity factor a(r,L) and the annualised costs for the technologies.
Output: Table (0.5)

With $r = 7\%$ and the following equations for the annuity factor $a(r, L)$ and the annualised costs c_{inv}

$$a(r, L) = \frac{r(1 + r)^L}{(1 + r)^L - 1}$$

$$c_{\text{inv}} = \text{overnight} \cdot a(r, L) + \text{FOM}$$

we receive these values for the different technologies:

Component	Kind	$c_{\text{var,other}}$	Overnight Cost	FOM	Lifetime	Efficiency	Annuity Factor	c_{inv}
Wind	power	1.62	1,286,467	15,148	30	–	0.08	118,819.75
Solar	power	0.00	367,867	9,304	40	–	0.08	36,897.39
H_2 electrolyzer	charge power	0.00	1,257,335	50,293	25	0.70	0.09	158,185.57
H_2 fuel cell	discharge power	0.00	1,068,642	53,432	10	0.50	0.14	205,582.58
H_2 storage	energy	0.00	1,604	0	100	–	0.07	112.41
Battery inverter	power	0.00	80,223	722	10	0.96	0.14	12,143.95
Battery storage	energy	0.00	100,279	0	30	–	0.08	8,081.12
Line	power	0.00	1,544,274	23,164	40	–	0.08	138,998.66

Table 1: Annuity factor and annualised costs for the technologies.

(b) In the no-line case, what does the dispatch of solar and wind look like in each node?
Output: Dispatch plot (0.5)

To find the optimal solution in the no-line case we minimize the objective function subject to the constraints and receive with the dual-simplex algorithm the value of

$$\min z = 3\,412\,846\,097.4\,€$$

The optimal dispatch results for the no-line case for solar and wind in each node are shown in Figure 1. A second output in Figure 2 illustrates how storage systems participate in the dispatch pattern.

It can be observed that wind and solar are combined only in the northern node, whereas the southern node instead uses a combination of solar power and storage. This is due to several reasons. In scenario B (no line) each node has to cover its own demand, so the dispatch follows the local resource.
Generation dispatch:
In the north the supply is a mix of wind and solar. Wind carries the base supply across day and night and dominates in winter, while solar adds generation during the daytime hours, so the dispatch stays relatively continuous. In the south no wind is built at all, so solar is the only source. Its dispatch is strongly diurnal: high peaks around midday and zero at night, with large swings between the seasons.
Storage dispatch:
Because the north already has a fairly balanced wind-solar mix, it needs less storage. The south depends entirely on solar, which produces nothing at night, so a lot of energy has to be shifted from day to night. That is why both the battery and the H_2 charge/discharge power and storage capacity are clearly higher in the south. Batteries handle the short day-night cycle, while H_2 covers the longer, seasonal shifts. Overall this split is the solver's least-cost choice: it follows the local production potential (north = windiest node, south = sunniest node) together with the technology costs.

(c) Compare the two scenarios. How do the investments into generation, storage, and transmission change?
Output: Show the investment costs in a table and plot the capacities.(1)

The following Table 2 shows the optimal solution for scenarios A and B, as well as savings through line inclusion (which is the cost of B - A) and the build line capacity in scenario A.

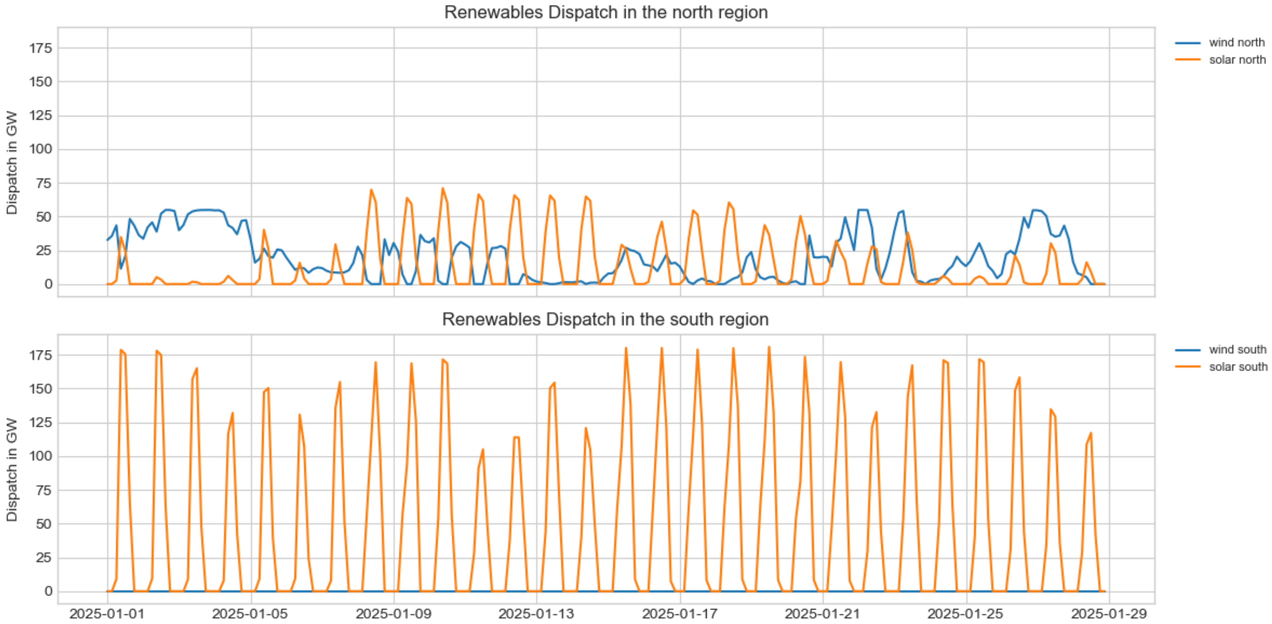

Figure 1: Dispatch of solar and wind in each node in the no-line case.

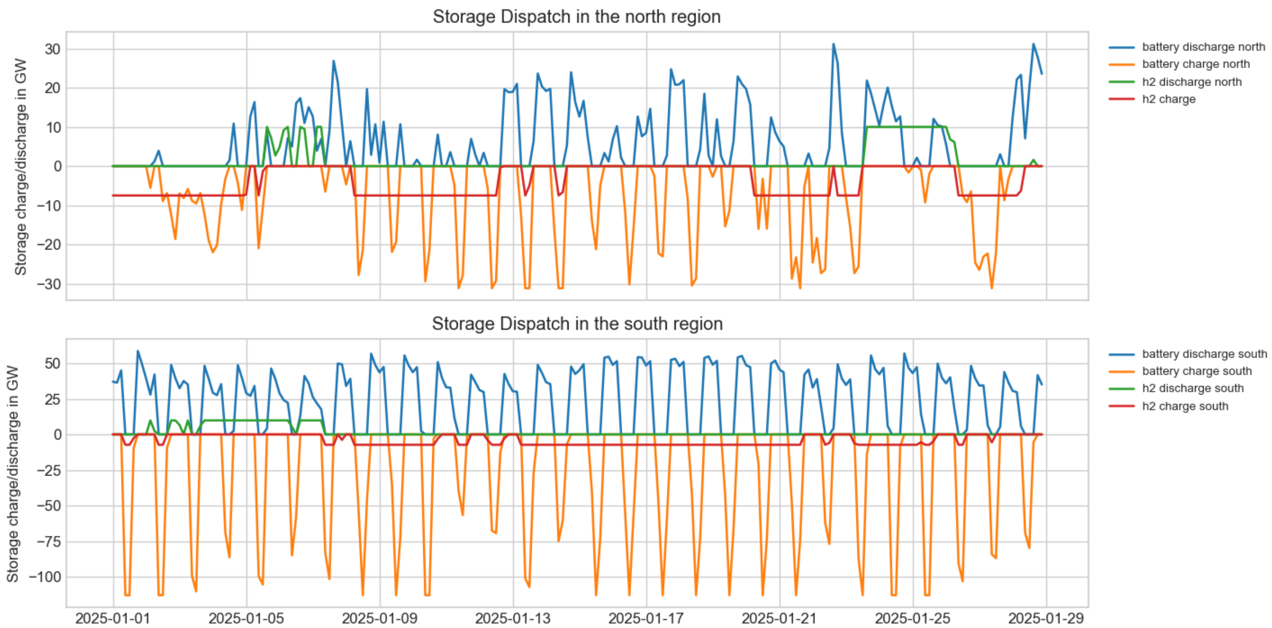

Figure 2: Participation of storage systems in the dispatch pattern in each node

TOTEX Scenario A (line)	3 056 740 408.5 €
TOTEX Scenario B (no line)	3 412 846 097.4 €
Savings through line inclusion	356 105 688.9 €
Build line capacity	14.06 GW

Table 2: Optimal solution for scenarios A and B – investment cost and build line capacity.

Furthermore, Table 3 compares the annual costs of each technology across both scenarios. The third column in the table indicates whether a technology is more expensive in the line or no-line case. Positive values indicate higher annual costs in scenario A (line), whereas negative values indicate higher annual costs in scenario B (no line).

Component	A (line) [Mio€/a]	B (no line) [Mio€/a]	Δ A–B [Mio€/a]
Wind	8 385	6 858	1 527
Solar	15 714	19 105	-3 391
H_2 electrolyzer	859	2 343	-1 484
H_2 fuel cell	1 824	4 078	-2 254
H_2 storage	167	359	-192
Battery inverter	1 536	1 751	-215
Battery storage	9 085	9 714	-629
Line	1 954	0	1 954

Table 3: Annual costs comparison between scenarios A and B for every technology.

Figure 3 shows the comparison of the power and energy capacity in each node and for either scenarios.

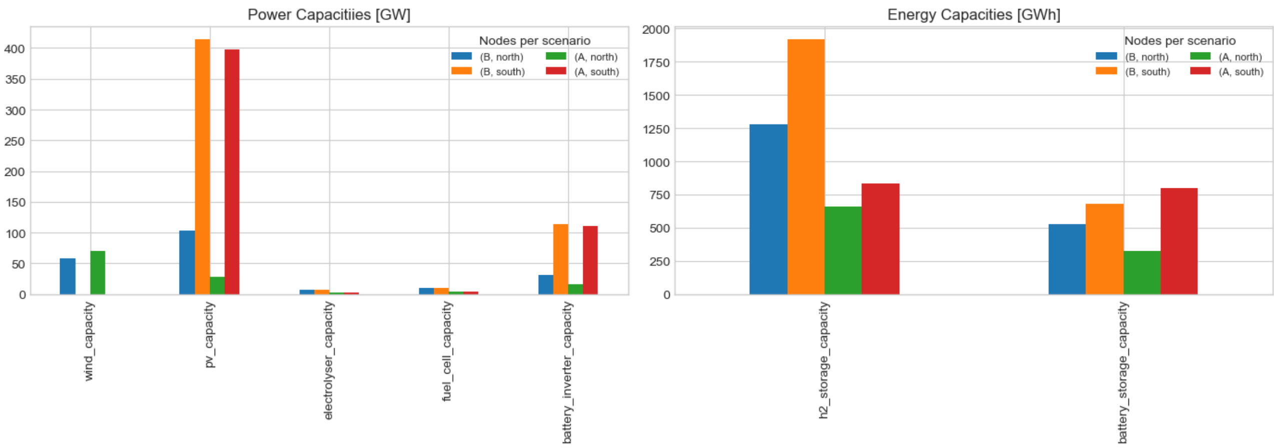

Figure 3: Power and energy capacity for each node and scenario.

The total investment costs shrink when line capacity is built (TOTEX falls from 3.41 to 3.06 billion €). With the line the regions no longer have to cover their demand on their own, so they can specialise.
Generation:
The north expands wind and strongly cuts its solar, turning into a cheap wind exporter, while the south stays solar-based. Wind in the north is favourable because its good resource potential combined with the costs helps to reduce total costs. The south keeps its strong solar potential and only needs to reduce its storage and a little of its production capacity to meet demand.
Storage:
Because excess energy can be transferred between regions instead of being stored locally, less storage is needed overall. This is mostly visible in the H_2 storage (it drops by roughly half) and in the northern battery, while the southern battery even grows slightly.
Transmission:

Line capacity is built to carry the energy and to smooth the supply across seasonal and day/night differences between the windy north and the sunny south.

(d) How does curtailment by technology and region change from the no-line case to the transmission case and why?
Output: Quantify and explain. (0.5)

The curtailment of wind and solar generation in the respective regions is shown in Figure 4 and Figure 5. Figure 4 illustrates the curtailment over time for each scenario, while Figure 5 presents the corresponding values aggregated over time and disaggregated by technology and node. Table 4 provides the numerical values underlying the visualization in Figure 5.

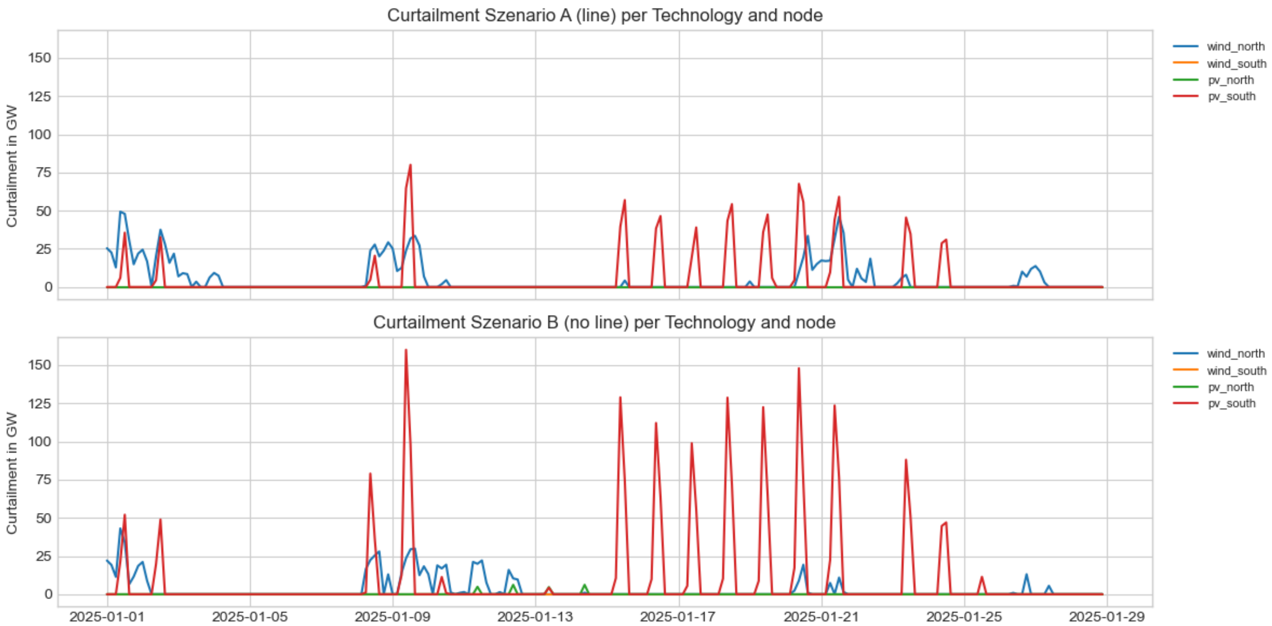

Figure 4: Times series of curtailment in each scenario.

Technology and Region	Scenario A [GWh]	Scenario B [GWh]
Wind north	2 044	3 376
Wind south	0	0
Solar north	66	0
Solar south	6 641	3 169

Table 4: Curtailment in GWh for each technology, region and scenario

Total curtailment falls when the line is built, but it does not fall everywhere.
South PV:
Curtailment drops strongly. In the no-line case the south has a large midday solar surplus that it cannot use or store, so it has to be spilled. With the line this surplus can be exported to the north instead of being curtailed.
North wind:
Curtailment actually increases. With the line the north overbuilds cheap wind to export it. During high-wind hours the line and storage cannot always absorb all of it, so a larger share of the wind potential is spilled.
North PV:
Drops to zero because the north barely builds any solar anymore in the line scenario. So the line reduces overall curtailment by letting the regions share their surplus, but it shifts the remaining curtailment towards the technology that is deliberately overbuilt for export (northern wind).

(e) How do the residual load curves change across the two scenarios?
Output: Plot the residual load curves and explain the differences. (0.5)

Figure 6 illustrates the residual load curve for the line and no-line cases.

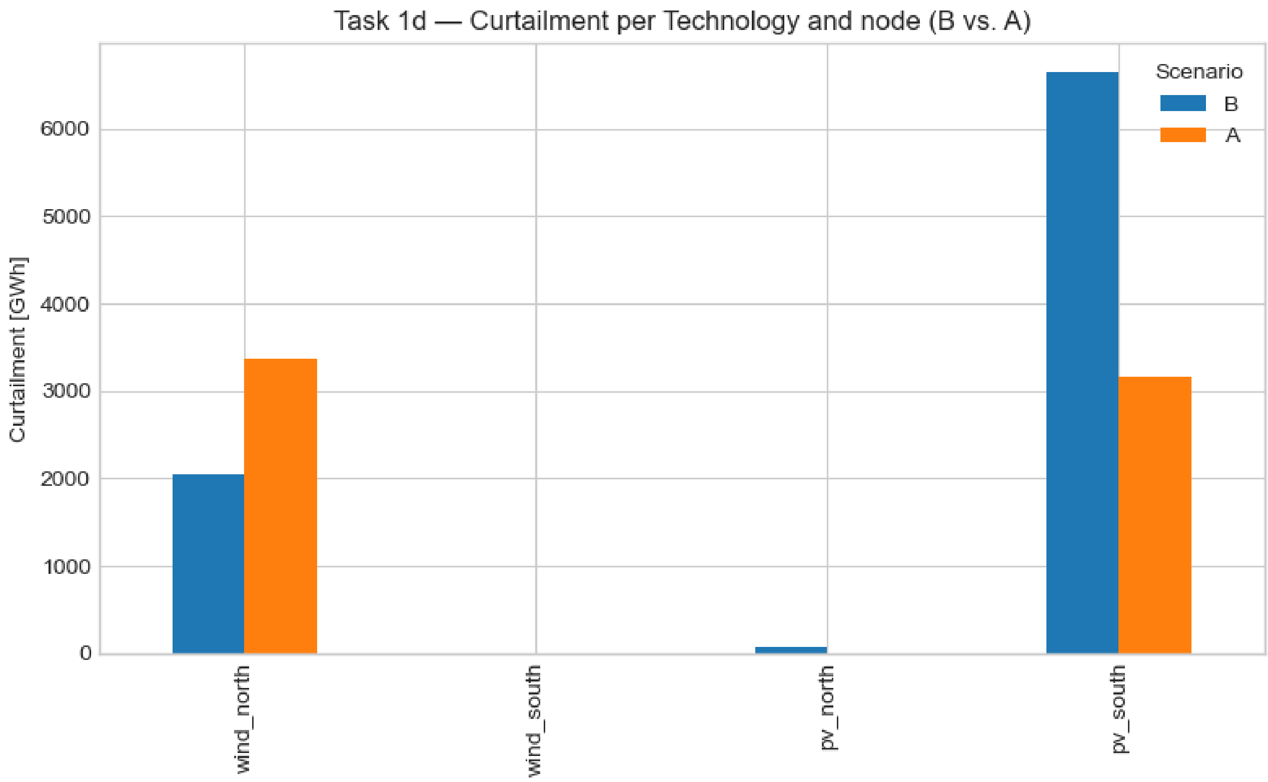

Figure 5: Curtailment in GWh for each technology, region and scenario.

The residual load is defined as the difference between electricity demand and dispatched renewable generation, aggregated over both nodes and sorted into a duration curve:

$$R(t) = D(t) - G_{\text{RES}}(t)$$

where $R(t)$ denotes the residual load, $D(t)$ the total demand, and $G_{\text{RES}}(t)$ the dispatched renewable generation.

Positive values of $R(t)$ represent residual demand that must be covered by storage, while negative values indicate surplus renewable generation that is absorbed by storage.

The peak of the curve is almost identical in both scenarios. This corresponds to the system-wide situation in which neither wind nor PV generation is available. When both regions simultaneously experience low wind and solar availability, the transmission line cannot provide additional support, since no surplus energy is available for spatial balancing. Consequently, storage must cover the demand. The main difference appears at the negative (surplus) end of the curve: with the transmission line, the curve is significantly flatter. The line enables spatial balancing of non-simultaneous surpluses between regions (northern wind and southern solar), thereby reducing the amount of energy that needs to be shifted through storage. This is consistent with the results in Sections 1c and 1d: with transmission, the system relies less on storage (resulting in lower hydrogen capacities) and experiences less curtailment, as spatial exchange partially substitutes temporal storage.

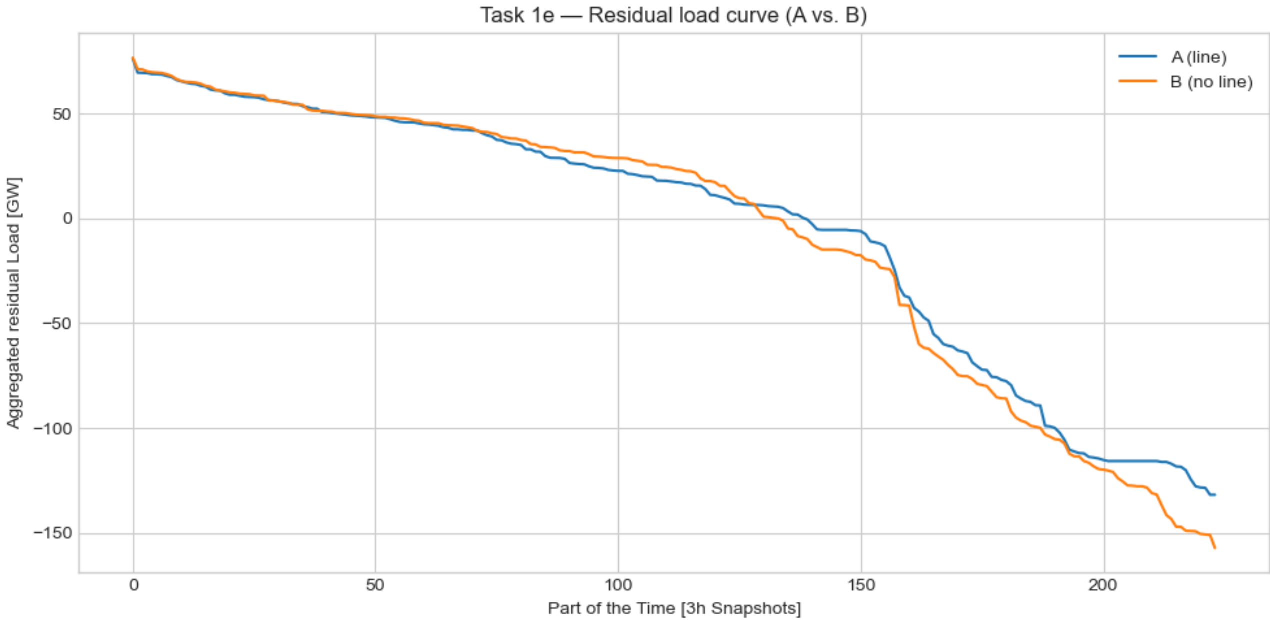

Figure 6: Residual load curve of scenarios A and B.

2 Task 2 - Questions on dual solution

(a) For each scenario, extract the shadow variable to the nodal energy balances and weight it by the nodal demand in the north and south, respectively. What do they represent? Why are they different in the two regions? When do you observe the highest values and why? (0.5)

The nodal electricity price at each node n and time step t is derived from the dual variable $\mu_n(t)$ of the power balance constraint, divided by the time resolution $\Delta t = 3\,\text{h}$:

$$\lambda_n(t) = \frac{\mu_n(t)}{\Delta t} \qquad \left[\frac{\text{EUR}}{\text{MWh}}\right]$$

The demand-weighted average price for each region is then computed as:

$$\bar{\lambda}_n = \frac{\sum_t \lambda_n(t) \cdot d_n(t)}{\sum_t d_n(t)}$$

Figure 7 shows the sorted nodal price duration curves for both scenarios and both regions.

The price duration curve shows the nodal price for each share of the time, sorted from high to low. For example, in roughly the most expensive fifth of the hours the price is above $100\,\frac{\text{EUR}}{\text{MWh}}$. The dual variable of the nodal energy balance represents the marginal cost of supplying one additional MWh at the respective node, i.e. the local electricity price ($\frac{\text{EUR}}{\text{MWh}}$). Weighting it by the nodal demand gives the average price paid by consumers in that region over the modelled period.
The prices differ between the two regions because each node draws on a different local resource mix. Scarcity hours are about equally frequent in both regions, the difference is at the low end. In scenario B (no line), the regions are fully decoupled and each forms its own price independently. The south relies exclusively on solar generation, which produces large midday surpluses at near-zero marginal cost. During these hours the price drops to almost zero, pulling the demand-weighted average down significantly. The north has a tighter wind-solar mix with almost no surplus hours, so a technology with positive variable cost (wind, at $1.62\,\frac{\text{EUR}}{\text{MWh}}$) sets the price during most time steps, resulting in a higher average. In scenario A (line), the two prices are coupled

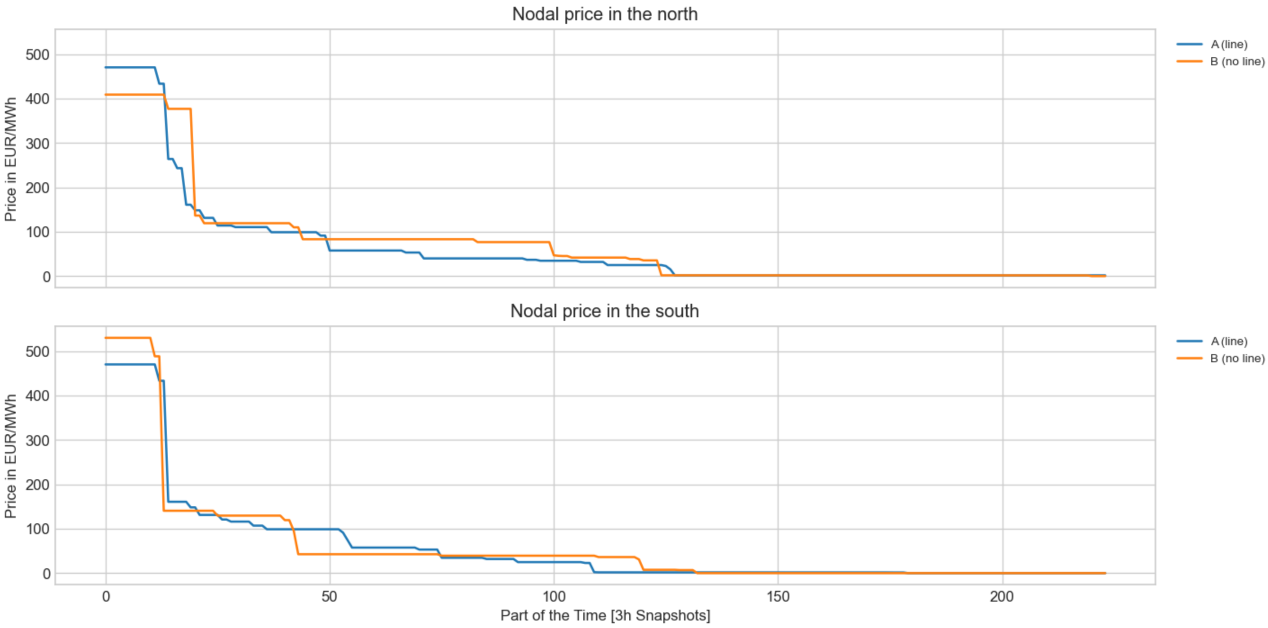

Figure 7: Nodal price duration curves for scenarios A (line) and B (no line) in each region.

during hours in which the line operates at full capacity, which flattens the duration curves and brings the regional averages closer together.
The highest price values are observed during dark-lull hours when neither wind nor solar generation is available. In these scarcity periods the only remaining dispatchable source is the H_2 fuel cell, whose dispatch sets the marginal price. In scenario B the extreme price peaks occur in the south, as a solar-only region it is most exposed when there is no sunshine. In scenario A the transmission line couples both nodes during scarcity, so both regions reach the same peak price simultaneously (the southern extreme is lowered and the northern peak is raised until they converge).

(b) In the transmission scenario, consumers of which region benefit more from the transmission line: the north, the south, or both equally? Briefly explain. (0.5)

Figure 8 compares the demand-weighted average nodal prices for both regions across the two scenarios.

Consumers in the north benefit more from the transmission line, although both regions gain. The north started with the higher demand-weighted price in scenario B, because its tighter wind-solar mix left it with almost no zero-price hours. The line grants the north access to the south's abundant midday solar surplus and allows it to avoid building as much additional PV and storage capacity, resulting in a larger absolute price reduction.
The south also benefits as it can export its surplus generation instead of curtailing it and it no longer needs to oversize storage to cover nighttime demand solely from local sources. The line thus balances the two nodal prices and reduces them in both regions, but to a greater extent in the north.

(c) Extract the shadow price of the line capacity constraint in the transmission scenario. What does it represent? In the scenario where the transmission capacity is zero, you can also extract the dual variable. Comparing the two, what does it say about the value of interconnection? (0.5)

The shadow price of the line capacity constraint is obtained by summing the duals of the upper and lower transmission flow constraints over all time steps. Figure 9 illustrates the comparison between the two scenarios and Table 5 reports the corresponding numerical values.

Scenario	Shadow price [$\frac{\text{kEUR}}{\text{MW}}$]
A (line)	≈ -11
B (no line)	≈ -59

Table 5: Shadow price of the line capacity constraint for both scenarios.

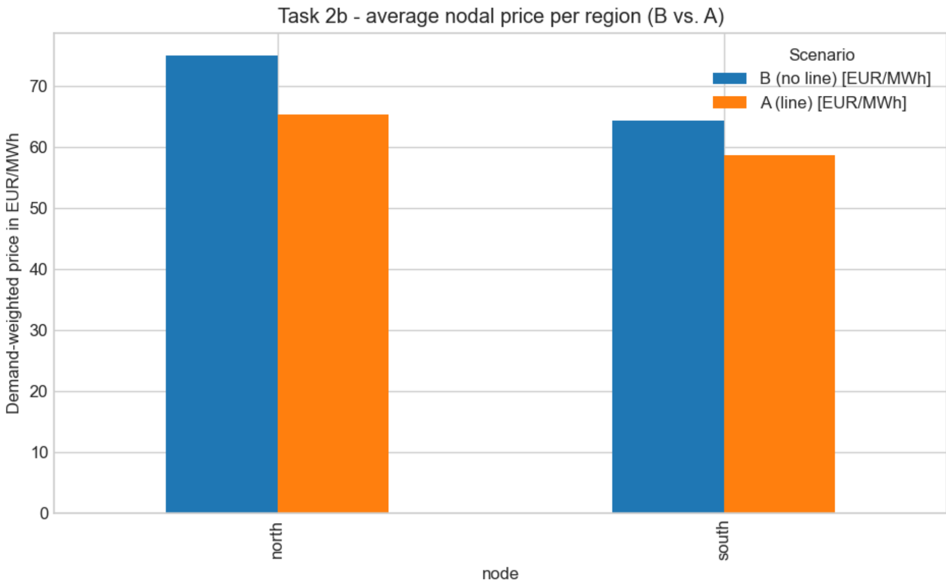

Figure 8: Demand-weighted average nodal price per region in scenarios A (line) and B (no line).

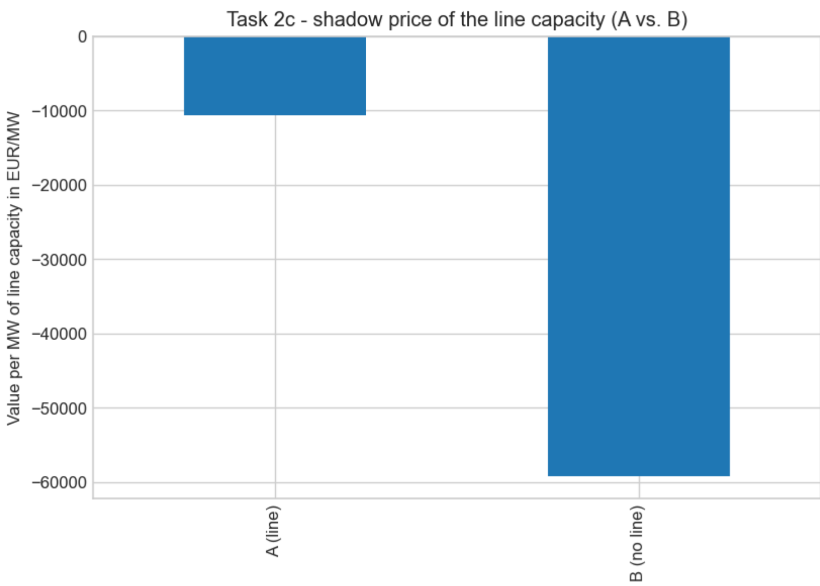

Figure 9: Shadow price of the line capacity constraint in scenarios A (line) and B (no line).

The shadow price represents the congestion rent: how much the total system cost would drop if the line could carry one more MW. It corresponds to the price difference between north and south in the hours when the line is operating at its capacity limit.
In scenario A (line), the shadow price is small (about $-11\,\frac{\text{kEUR}}{\text{MW}}$). Since the model is free to optimise the line capacity, it expands the line until the marginal benefit of one additional MW equals its annualised investment cost, so the residual marginal value is close to zero by construction.
In scenario B (no line), the line capacity is forced to zero. The no-line constraint reflects the value of the very first MW of interconnection (about $-59\,\frac{\text{kEUR}}{\text{MW}}$). This value is roughly 5.6 times larger than the marginal value in scenario A, illustrating strongly decreasing returns to interconnection. Interconnection is highly valuable, but with clearly decreasing value. The first connections yield the greatest cost savings and additional capacity is built only as long as its marginal benefit exceeds its marginal cost.

(d) Which advantages and disadvantages of heavy reliance on cross-regional interconnection does this stylised model not capture? No calculation needed. (0.5)

The model captures the cost minimising value of interconnection under idealised conditions, but neglects several important real world effects.
Disadvantages not captured:
Highly interconnected systems are susceptible to cascade failures. A fault on one line can overload neighbouring lines and propagate failures across the entire network. The model assumes lossless, perfectly controllable flows and therefore cannot represent this systemic risk. Operating a highly meshed grid also requires close coordination between system operators that may have different regulatory frameworks, pricing zones, and technical standards (coordination costs and the resulting decision-making delays are not modelled). Cross-border systems are furthermore exposed to political risk. Unilateral decisions by a neighbouring country, such as export restrictions or unplanned capacity reductions, can leave an interconnected region undersupplied. Large-scale overhead transmission lines also face public acceptance challenges (e.g. SuedLink in Germany), which can delay or entirely prevent construction. Finally, rising interconnection can increase local curtailment for specific technologies if the line redirects flows in a way that reduces the local value of that generation, which can affect the profitability of individual projects.
Advantages not captured:
While the model captures the direct investment cost reduction, it understates several further benefits. In practice, transmission lines improve system security and can prevent local blackouts by importing energy from regions with available supply during local shortfalls. In competitive electricity markets, interconnection reduces local market power and promotes price convergence, leading to more efficient dispatch. The stylised two-node representation also ignores full network topology, so the value of redundancy and the benefits of a meshed grid (such as N-1 security and reactive power support) are not reflected. Overall, the model provides a partial view of the true social value of interconnection, since it omits both the non-market benefits and the systemic risks that become relevant at high integration levels.